

RETRACTION — Keeper’s “a substrate gauge symmetry acts by Jordan automorphisms, hence fixes e ” principle is REFUTED (Cal’s $SO(5)$ counterexample); the muon banks on the $n=e$ PROOF, not a principle

Keeper

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K983 — The principle is refuted; compute the thing it would have assumed

What I floated (and now retract)

K982 offered a fork: bank the muon Derived by ruling forcing the principle “a gauge symmetry of the substrate acts by Jordan automorphisms of the substrate algebra, hence fixes the cone identity e .” I recommended leaning to accept it.

Cal’s refutation (clean counterexample)

$SO(5)$ is itself a BST symmetry — the electroweak group descends from it — and $SO(5)$ does NOT act by Jordan automorphisms of the spin factor. A generic $SO(5)$ rotation moves e (the vector 5 is irreducible under $SO(5)$; e is not $SO(5)$ -invariant); only the $SO(4)$ subgroup that fixes e acts by Jordan automorphisms. So “it is a symmetry” does not imply “it fixes e .” The principle, applied to the electroweak $SO(4)$, *assumes* it is the e -fixing conjugate — which is exactly the open question ($n=e$). Stamping Derived on it is the favored-basis wave-through we refuse ([feedback_no_wave_through_on_a_perfect_number])). Retracted.

The lesson (folds into the standing discipline)

A “principle” that would bank a claim is itself a claim — and if the principle *assumes* the very identification in question (here: that the physical subgroup is the canonical/structure-preserving one), it is a favored-basis assumption wearing a general-principle costume. Test any bank-enabling principle against the theory’s OWN structure for a counterexample *before* invoking it ($SO(5)$ was the counterexample, sitting in plain sight). Same class as: object-location $\nrightarrow$ object-innocence (K969), a clean form is a candidate not a bank (K942-lineage), no wave-through on a perfect number (K981-lineage). The auditor floating a wave-through, and the referee catching it, is the machine working — this time on me.

The corrected path (single, clean)

The muon banks DERIVED by PROOF, not principle: - (i) Cal — spinor-local $n=e$: derive which vector the $Sp(2)$ -spinor’s $Spin(4)$ actually fixes, target-innocently (from the rep structure, not by matching), and check it = the cone identity e . This is the vacuum-alignment / misalignment-angle-zero computation — a day, separated from the deep $\dim-11 \rightarrow 12$ embedding (Cal §127). - (ii) Lyra — the independent second structural route: does the complex structure $J = SO(2) = U(1)_Y$ force $n=e$ (F633)? If it lands independently of Cal’s rep route $\rightarrow$ two structural routes = Derived clean (the (b) clause), no principle needed. Keeper fires K967 the moment $n=e$ derives target-innocently. K982 fork-(ii) is superseded by this note.

— K983, Keeper, 2026-07-29. Retracts the “gauge symmetries fix e ” principle (SO(5) counterexample, Cal); the muon banks on the $n=e$ proof (Cal spinor-local) + Lyra’s complex-structure second route. See [[Keeper_K982_tier_rulings_TAU_Fitted_derived_final_via_rigorous_pole_MUON_at_the_Derived_doorstep_e_equals_f_separ 07-29]], Cal §127, team prompt 2026-07-29b.